AARAL AGRAWAL

B.Tech Computer Science Student
aaralagrawal7@gmail.com — +91 8107260988 — Tonk, Rajasthan
github.com/Aaral-Agrawal — linkedin.com/in/aaral-agrawal-4a3b99283

Summary

AI/ML-focused 3rd-year B.Tech Computer Science student with strong interest in machine learning, NLP, intelligent systems, and research-driven problem solving. Experienced in building AI-powered applications involving LLMs, speech analysis, NLP evaluation systems, and data extraction tools. Skilled in Python, JavaScript, React.js, and problem-solving with a passion for applying AI to real-world challenges and exploring advanced research in Artificial Intelligence and Data Science. Interested in exploring research problems related to intelligent systems, generative AI, and human-centered AI applications.

Education

B.Tech in Computer Science — Banasthali Vidyapeeth, Rajasthan 2023 – 2027
CGPA: 8.9/10
Relevant Coursework: DSA, Operating Systems, DBMS, Algorithms, Computer Networks

Experience

Hackathon Developer — Hack Celestia 2024

- Built an AI-based solution within 48 hours, achieving 2nd place among 50+ competing teams.
- Contributed to system design, integration, and final presentation.

Projects

VokMock – AI-Powered VR Interview Platform — GitHub *Three.js, WebXR,MongoDB, OpenAI LLM*

- Developed an AI-powered VR interview simulation platform integrating LLMs and speech analysis for adaptive mock interview experiences.
- Implemented OpenAI APIs and Whisper-based speech-to-text processing for automated feedback generation across 20+ interview scenarios.

RAAGAX:Music Note Detection Model — GitHub *Python, ML*

- Designed and trained a machine learning model for real-time musical note detection and visualization using audio signal processing.
- Achieved 85% detection accuracy while significantly reducing manual transcription effort.

HVAL – Human Evaluation of Machine Translation System — GitHub *MERN Stack, NLP, JWT, Chart.js*

- Built an NLP-focused platform for evaluating machine translation quality using human-centered scoring metrics.
- Implemented analytical dashboards, dataset management, and role-based workflows for evaluators and researchers.

Financial Data Extractor App — GitHub *Python, Streamlit, OpenAI*

- Developed an AI-assisted financial insight extraction system using OpenAI models to analyze financial articles.
- Designed an interactive Streamlit dashboard that

Technical Skills

Programming Languages: Python, C++, JavaScript, SQL
AI/ML & Data: Machine Learning Basics, Natural Language Processing (NLP), OpenAI APIs, Speech-to-Text Systems, Data Analysis
Web Technologies: React.js, MERN Stack, HTML5, CSS3, Streamlit
Libraries & Tools: Chart.js, Three.js, Leaflet.js, Git, GitHub, VS Code
Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS

Research Interests

Artificial Intelligence & Machine Learning • Natural Language Processing (NLP) • Generative AI & Large Language Models • Human-AI Interaction • Intelligent Interactive Systems • AI for Real-World Applications

Achievements

- 2nd place – Hack Celestia Hackathon for AI-based solution.